AWS Lab: Online EBS Volume Expansion & OS Filesystem Growth

Duration : 60 Min.

Scenario

A production EC2 Linux server's EBS root volume (/dev/nvme0n1) is at 98% capacity. The cloud administrator expanded the volume from 30 GB to 80 GB using the AWS CLI: 'aws ec2 modify-volume --volume-id vol-0123 --size 80'. However, 'df -h' inside the operating system still shows only 30 GB on the root partition. The engineer must use 'growpart' to extend partition 1 and 'resize2fs' / 'xfs_growfs' to expand the filesystem online with zero downtime.

Input Details

The cloud environment has been pre-configured with the following parameters: - **Platform:** Amazon Elastic Block Store (EBS) & Linux OS - **Block Device:** `/dev/nvme0n1` (expanded from 30 GB to 80 GB) - **Partition & Filesystem:** Partition 1 (`/dev/nvme0n1p1`), filesystem online growth

Username & Naming Conventions

Your candidate prefix is: **`<prefix>`** (derived from your login email before the `@` symbol, e.g. `student-demo01`). All custom resources and cloud configurations should incorporate this convention: - **Resource Naming:** Prepend or append `<prefix>` to your created AWS resources (e.g. `<prefix>-resource`). - **Candidate Tag:** Add tag `CandidatePrefix` = `<prefix>` on all managed cloud resources.

Task Objectives

Perform the following technical objectives directly in the AWS Management Console or using the AWS CLI: 1. Inspect disk and partition geometry using lsblk. 2. Execute growpart /dev/nvme0n1 1 to extend the partition. 3. Execute the appropriate filesystem resize utility (resize2fs or xfs_growfs) based on filesystem type. 4. Verify with df -h that usable space reflects 80 GB, documenting execution in ~/ebs_resize.log.

Verification

Once completed, run the verification script to check your progress and receive your score. The verification system will inspect the live resources in the AWS cloud:

```
python run.py
```

Grading Criteria

Your performance will be evaluated based on the following test cases:

Test Case	Requirement	Marks
TC1	Geometry inspection: identify disparity between 80 GB raw block device and 30 GB partition	5 Marks
TC2	Partition extension: growpart successfully expands partition 1 without data corruption	5 Marks
TC3	Online filesystem growth: filesystem successfully resized online without reboot	5 Marks
TC4	Verification: df confirms 80 GB capacity and test file creation succeeds	5 Marks

Total Score: 20 Marks

Important Notes

- Perform all actions directly in the AWS Management Console or using the AWS CLI.
- Ensure all resource configurations, tags, and naming conventions match the requirements.
- Do not terminate or alter pre-existing cloud infrastructure outside the scope of this assessment.